

Interpretable Molecular Property Predictions Using Marginalized Graph Kernels

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Cite This: *J. Chem. Inf. Model.* 2023, 63, 4633–4640



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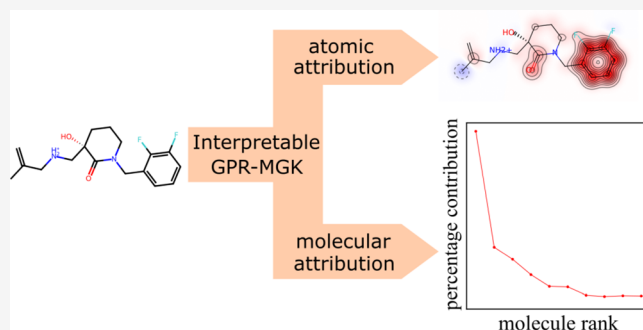


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ABSTRACT: Marginalized graph kernels have shown competitive performance in molecular machine learning tasks but currently lack measures of interpretability, which are important to improve trust in the models, detect biases, and inform molecular optimization campaigns. We here conceive and implement two interpretability measures for Gaussian process regression using a marginalized graph kernel (GPR-MGK) to quantify (1) the contribution of specific training data to the prediction and (2) the contribution of specific nodes of the graph to the prediction. We demonstrate the applicability of these interpretability measures for molecular property prediction. We compare GPR-MGK to graph neural networks on four logic and two real-world toxicology data sets and find that the atomic attribution of GPR-MGK generally outperforms the atomic attribution of graph neural networks. We also perform a detailed molecular attribution analysis using the FreeSolv data set, showing how molecules in the training set influence machine learning predictions and why Morgan fingerprints perform poorly on this data set. This is the first systematic examination of the interpretability of GPR-MGK and thereby is an important step in the further maturation of marginalized graph kernel methods for interpretable molecular predictions.



1. INTRODUCTION

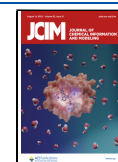
Molecular machine learning is increasingly employed to predict the physical, chemical, and biological properties of small molecules. Most molecular machine learning workflows require a separate function that creates a fixed numerical description of molecules to serve as input to a machine learning algorithm, for example, using molecular fingerprints or computed physicochemical properties. While such workflows have led to successful applications in drug discovery, drug delivery, organic chemistry, and crop science,^{1–6} identification of the “correct” descriptor requires domain knowledge or needs to be manually optimized. This process can be time- and resource-consuming and, more importantly, can lead to poor performance if using a suboptimal descriptor. More recently, major research has investigated machine learning architectures that directly accept molecular structures as input, e.g., using SMILES encodings or by using molecular graphs. Graph neural networks (GNNs) are becoming increasingly popular because they can learn from the graph structures of molecules.⁷ Similarly, the marginalized graph kernel (MGK) was originally proposed by Kashima two decades ago.⁸ However, it has not yet gained widespread popularity in molecular machine learning due to its tremendous computational cost that limited deployment. Recently, Tang et al. developed a high-throughput solver for MGK using graphical processing units (GPUs), which increases the computational speed of MGK by 3 to 4 orders of magnitude. This has made MGK a practically feasible

machine learning method that can be applied to real-world problems of thousands of drug-like molecules.⁹ For example, we and others have recently employed marginalized graph kernels as input to Gaussian Process (GP) models, which shows promising results for predicting quantum mechanical¹⁰ and thermodynamic¹¹ properties and on a range of benchmark data sets spanning drug discovery and toxicity.¹² Importantly, MGK enables competitive prediction accuracy compared to popular directed message-passing neural networks (D-MPNN),¹³ but with improved uncertainty quantification of predictions.¹²

In spite of their promise, a major drawback of models that learn directly from the molecular structure is the lack of interpretability. While many classical algorithms and descriptors enable some form of correlating predictions to certain molecular properties or substructures for example using SHapley Additive exPlanations (SHAP) and similar approaches,^{14,15} this is not inherently true for most representation-learning algorithms. Interpretability of the predictive

Received: March 13, 2023

Published: July 28, 2023



architecture is important to detect biases, avoid overfitting, increase trust in predictive models, and inform human scientists to accelerate and improve molecular reasoning and optimization. Accordingly, multiple approaches have been developed to enable the interpretation of GNNs, including class activation map (CAM),¹⁶ integrated gradient (IG),^{17,18} GradCAM,¹⁹ GraphLime,²⁰ PGM-explainer,²¹ GNNExplainer,²² improving explainability with orthonormalization and induced sparsity,²³ and several studies have benchmarked these methods.^{24–27} Most of these studies focus on atomic attribution (also known as “graph attributions”²⁴ or “heat maps”¹⁵); i.e., for a molecule to be predicted, these algorithms identify the atoms or molecular substructures of this molecule that are considered by the machine learning algorithm to reach a specific prediction.

To date, less research has been conducted on the interpretability of graph kernels.^{28,29} We postulate that the deterministic functional form of MGK should enable the derivation of interpretations. Specifically, we here derive two classes of interpretability for the Gaussian process regression marginalized graph kernel (GPR-MGK):

- Atomic attribution: attribute the prediction to the contribution of atoms in the predicted molecule.
- Molecular attribution: attribute the prediction to the contribution of other molecules in the training set.

Both interpretations retain additivity in that a prediction is equal to the sum of all molecular or atomic contributions. This contrasts with the explainability of GNNs, which commonly violates additivity. We evaluate the performance of our atomic attribution with established benchmarks using substructure-based logic data sets and compare our results with those achieved by Lengeling et al. using GNNs.²⁴ To assess our molecular attribution approach, we use the FreeSolv data set and compared the results of GPR-MGK with GPR-RBF-Morgan using the established radial basis function kernel on Morgan fingerprints.

II. METHODS AND MATERIALS

Gaussian Process Regression. Gaussian process regression (GPR)³⁰ is a nonparametric Bayesian model. Given n molecules $\mathbf{X}_T$ and target property $\mathbf{y}_T$ as a training set, the prediction of a GPR on a test set $\mathbf{X}_t$ of m molecules is defined as a multivariate Gaussian distribution $N(\mathbf{y}_\nu, \Sigma_t)$ with parameters

$$\mathbf{y}_t = \mathbf{K}_{Tt}^T \mathbf{K}_{TT}^{-1} \mathbf{y}_T \quad (1)$$

$$\Sigma_t = \mathbf{K}_{tt} - \mathbf{K}_{Tt}^T \mathbf{K}_{TT}^{-1} \mathbf{K}_{Tt} \quad (2)$$

where $\mathbf{K}_{TT} = K(\mathbf{X}_T, \mathbf{X}_T) \in \mathbb{R}^{n \times n}$, $\mathbf{K}_{Tt} = K(\mathbf{X}_T, \mathbf{X}_t) \in \mathbb{R}^{n \times m}$, $\mathbf{K}_{tt} = K(\mathbf{X}_t, \mathbf{X}_t) \in \mathbb{R}^{m \times m}$ are kernel matrices that define the similarity between the molecules within and between the training set $\mathbf{X}_T$ and the test set $\mathbf{X}_t$. For Morgan fingerprint representation, we use the established radial basis function kernel (RBF)

$$K_{\text{RBF}}(\mathbf{X}, \mathbf{X}') = \exp\left(-\frac{\|\mathbf{X} - \mathbf{X}'\|^2}{2\sigma^2}\right) \quad (3)$$

where σ is a length scale hyperparameter. For graph representations, we will use marginalized graph kernels, as described in the next section.

Marginalized Graph Kernel. A marginalized graph kernel (MGK)⁸ computes the similarity between a pair of (molecular) graphs by performing all possible random walks on the graphs simultaneously and summing the similarities of two paths multiplied by their probability across all path pairings (path kernels):

$$K(G, G') = \sum_{l=1}^{\infty} \sum_{\mathbf{h}} \sum_{\mathbf{h}'} \left[\left(\prod_{i=2}^l p_t(h_i | h_{i-1}) \right) \left(\prod_{j=2}^l p'_t(h'_j | h'_{j-1}) \right) \times \left(\prod_{k=2}^l \kappa_v(v_{h_k}, v_{h'_k}) \kappa_e(e_{h_k h_{k-1}}, e'_{h'_k h'_{k-1}}) \right) \right] \quad (4)$$

where G and G' are two input graphs to be compared; $\mathbf{h}$ and $\mathbf{h}'$ are random walk paths of length l ; and p_s, p_q, p_t are the starting probability, stopping probability, and transition probability of the random walks, respectively. In practice, instead of computing eq 4 by explicit random walk sampling, this equation can be reformulated into a linear system in a matrix form which is then efficiently computed with GPUs using the GraphDot package to calculate the analytical solution of the series convergence.³¹ For a more in-depth understanding of the mathematical details involved, we refer the reader to references 8–10.

In this study, we set the starting probability distribution to be discrete uniform

$$p_s(h) = \frac{1}{N(G)}, h \in G \quad (5)$$

where $N(G)$ is the number of nodes in graph G . The stopping probability p_q is set to a constant number for all atoms and is a hyperparameter of the model, and the transition probability is again a discrete uniform distribution set to $1/n_{\text{neighbor}}$ (equal probability to transition to any of the neighboring atoms during the random walk).

$\kappa_v(\bullet, \bullet)$ is the atom kernel that computes the similarity between a pair of atoms. It is computed using a weighted, normalized sum

$$\kappa_v(v, v') = \frac{\sum_i c_i \mu_i(\phi_i(v), \phi_i(v'))}{\sum_i c_i} \quad (6)$$

where μ_i is the feature kernel for the i -th atom feature ϕ_i and the weight hyperparameter c_i that determines the importance of ϕ_i . Similarly, $\kappa_e(\bullet, \bullet)$ is the bond kernel that computes the similarity between a pair of bonds, which is defined similarly to the atom kernel. For details on the features of atoms, bonds, and associated kernel functions, we refer the reader to ref 12.

Of note, in previous work,^{11,12} Xiang et al. set the starting probability using a graph-size-dependent normalization to enhance model performance in cross validation

$$\bar{K}(G, G') = F \frac{K(G, G')}{\sqrt{K(G, G)K(G', G')}} \exp\left[-\frac{(K(G, G) - K(G', G'))^2}{\lambda^2}\right] \quad (7)$$

where F and λ are hyperparameters to be optimized. However, eq 7 violates the additivity of the graph kernel to graph nodes,

which, as demonstrated later, is a prerequisite for atomic attribution calculation. Hence, we introduce starting probability normalization (eq 5), which preserves the additivity property of MGK.

Atomic Attribution. Atomic attribution decomposes the GPR-MGK prediction to the sum of atomic contributions in the molecule from paths that start at this atom.

We rewrite eq 4 as

$$K(G, G') = \sum_{h_1 \in G} p_s(h_1) \sum_{l=1}^{\infty} \sum_{\mathbf{h}(h_1)} \sum_{\mathbf{h}'} \left[\left(\prod_{i=2}^l p_t(h_i|h_{i-1}) \right) \left(\prod_{j=2}^l p_t(h'_j|h'_{j-1}) \right) \times \left(\prod_{k=2}^l \kappa_v(v_{h_k}, v'_{h'_k}) \kappa_e(e_{h_k h_{k-1}}, e'_{h'_k h'_{k-1}}) \right) \right] \times K_a(h_1, G')$$

$$= \sum_{h_1 \in G} p_s(h_1) K_a(h_1, G') \quad (8)$$

where $\mathbf{h}(h_1)$ are all random walks that start at atom h_1 and $K_a(h_1, G')$ is the atom-molecule kernel that evaluates the similarity between the atom h_1 in molecule G (random walk across G only starts from atom $h_1 \in G$) and molecule G' (random walk starts from all atoms in graph G'). Considering single-target regression $\mathbf{y}_T \in \mathbb{R}^{n \times 1}$, we substituted eq 8 into eq 1 to obtain eq 9

$$\mathbf{y}_t = \mathbf{K}_{TT}^T \mathbf{K}_{TT}^{-1} \mathbf{y}_T$$

$$= \begin{pmatrix} \sum_{h_1 \in G_{t,1}} p_s(h_1) & \cdots & \vdots \\ K_a(h_1, G_1) & & \\ \vdots & \sum_{h_1 \in G_{t,i}} p_s(h_1) K_a(h_1, G_j) & \vdots \\ \vdots & \cdots & \sum_{h_1 \in G_{t,m}} p_s(h_1) \\ & & K_a(h_1, G_n) \end{pmatrix} \mathbf{B}$$

$$= \begin{pmatrix} \vdots \\ \sum_{h_1 \in G_{t,i}} \sum_{j=1}^n p_s(h_1) K_a(h_1, G_j) b_{j1} \\ \vdots \end{pmatrix} \quad (9)$$

where $b_{j1} = \sum_{k=1}^n (\mathbf{K}_{TT}^{-1})_{jk} (\mathbf{y}_T)_k$. For molecule $G_{t,i}$ in the test set, the atomic contribution of the atom $h_1 \in G_{t,i}$ is

$$\sum_{j=1}^n p_s(h_1) K_a(h_1, G_j) b_{j1} \quad (10)$$

The sum of the atomic contributions over all atoms is equal to the predicted value of GPR on the molecule.

Molecular Attribution. Molecular attribution decomposes the GPR prediction into the sum of contributions of a single data point in the training set. Considering single-target regression $\mathbf{y}_T \in \mathbb{R}^{n \times 1}$, we rewrite eqs 1 and 2 (the prediction of GPR) as

$$\mathbf{y}_t = \mathbf{K}_{Tt}^T \mathbf{K}_{TT}^{-1} \mathbf{y}_T = \mathbf{A} \mathbf{y}_T = \begin{pmatrix} \vdots \\ \sum_{j=1}^n a_{ij} y_j \\ \vdots \end{pmatrix}$$

$$\Sigma_t = \mathbf{K}_{tt} - \mathbf{A} \mathbf{K}_{Tt} \quad (12)$$

where $a_{ij} = \sum_{k=1}^n (\mathbf{K}_{Tt}^T)_{ik} (\mathbf{K}_{TT}^{-1})_{kj}$ is the percentage contribution of molecule j in the training set to the prediction of molecule i in the test set and $a_{ij} y_j$ is the absolute contribution to the predicted value. The sum of all absolute contributions is equal to the predicted value for molecule i in the test set of the GPR model. Note that molecular attribution has no requirement for the kernel function, so it can be used for any GPR model, including RBF kernels.

Hyperparameter Optimization. The hyperparameters of the MGK include variables such as the stopping probability and parameters to calculate atom and bond similarity. These must be optimized for a specific prediction task. We used an automated Bayesian optimization algorithm, the Tree of Parzen Estimators (TPE), to obtain optimal hyperparameters.^{32,33} Leave-one-out cross-validation (LOOCV) performance of a GPR can be efficiently computed from the inverse of the kernel matrix of the training data set³⁰

$$\mu_i = y_i - \frac{(\mathbf{K}_{TT}^{-1} \mathbf{y}_T)_i}{(\mathbf{K}_{TT}^{-1})_{ii}} \quad (13)$$

$$\sigma_i^2 = \frac{1}{(\mathbf{K}_{TT}^{-1})_{ii}} \quad (14)$$

where μ_i and σ_i^2 are the predicted mean and variance, respectively, of data point i . The deviation between the LOOCV prediction and the ground truth is the LOOCV loss, which is the objective function to be minimized by Bayesian optimization. We use the root-mean-square error (RMSE) as the metric for regression and the area under the receiver operating characteristic curve (ROC-AUC) for classification. Instead of performing LOOCV on the whole data set, Hyperopt-subsets-LOOCV randomly splits the data into multiple subsets and then performs LOOCV on each subset separately. The averaged LOOCV losses of the subsets are then the objective function for Bayesian optimization. We optimized subset size N_{subset} through grid search and found robust performance across a range of different N_{subset} values. We set $N_{\text{subset}} = 100$ given its good performance for both prediction and interpretability (Figures S1 and S2).

Graph Neural Networks. We employed GNNs as the reference model for atomic attribution task to contextualize the performance of GPR-MGK. We selected the top four GNN-explanation combinations from Sanchez-Lengeling et al.'s paper:²⁴ Message-Passing Neural Network with Class Activation Map (MPNN-CAM), Graph Convolutional Neural Network with Class Activation Map (GCN-CAM), Graph Attention Network with Class Activation Map (GAT-CAM), and Graph Convolutional Neural Network with Integrated Gradients (GCN-IG). The GNNs were configured as three-layer networks with 20-dimensional edge embeddings, 50-dimensional node embeddings, and a global size of 100. This architecture demonstrates nearly optimal predictive performance based on hyperparameter grid-search results.²⁴

Data Sets. The data sets used in this study are summarized in Table 1. The logic data sets are artificial data sets in which

Table 1. Datasets Used in This Paper

task	data set	description	no. of molecules (training)	no. of molecules (test)
atomic attribution	logic7	fluoride and carbonyl	3840	486
	logic8	unbranched alkane and carbonyl	7753	918
	logic10	amine and ether2 & benzene	7718	969
	benzene	benzene	10000	2000
	Liver toxicity	hepatotoxicity	291	66
	Ames	mutagenicity	5204	1302
molecular attribution	FreeSolv	solvation free energy	321	321

the label of a molecule is set as “positive” if a molecule contains specific chemical substructures and “negative” otherwise.¹⁸ Therefore, the molecules in the logic data sets have clear, binary ground-truth atomic attributions (atoms belonging to the desired substructure are “positive”, all other atoms are “negative”), which enables quantitative evaluation of the performance of different machine learning model interpretations. We used the same data sets and data splitting as Sanchez-Lengeling et al.²⁴ to directly contextualize the performance of our interpretability predictions. In addition, two real-world toxicology data sets were analyzed as curated by Rao et al.³⁴ The liver toxicity data set was created by Liu et al.³⁵ and contains annotations of 12 molecular substructures as critical to cause human liver injuries (“toxicophores”). Similarly, Hansen et al.³⁶ provided a data set on the potential of a molecule to cause mutagenicity based on the Ames test and included 46 molecular substructures labeled as toxicophores by Sushko et al.³⁷ All data sets are binary classification tasks.

To study the behavior of molecular attribution, we used the FreeSolv data set, a publicly available database capturing the hydration free energy of 642 molecules.³⁸ We carried out a 50:50 scaffold split to obtain the training set and the test set. In our scaffold split, molecules are divided into different subsets according to their Murcko scaffold³⁹ as calculated by RDKit.⁴⁰

The scaffold groups are sorted from highest to lowest according to the number of molecules sharing that scaffold and placed into the training set until the number of molecules reaches the desired training set size (i.e., cover half of the total data). The remaining molecules are placed into the test set.

III. RESULTS AND DISCUSSION

Atomic Attribution: Logic and Toxicology Data Sets.

We trained our GPR-MGK model on four logic and two real-world benchmark data sets. We observed strong predictive performance with classification ROC-AUC larger than 0.96 on the four logic data sets (Table S1, Figure S1), indicating that GPR-MGK models can learn the simple chemical relationships in these data sets efficiently and confirming previous results on the effectiveness of MGK for molecular machine learning. Real-world data sets are more challenging, as evidenced by a ROC-AUC score of 0.66 for the hepatotoxicity data set and score of 0.88 for the Ames mutagenicity data set. GPR-MGK and GNNs have similar predictive performance on all data sets.

We next investigated the interpretability of our machine learning method by assessing the performance of our model on atomic contributions tasks. Since the labels in the data set are assigned based on the presence or absence of certain functional groups, the ground truth of atomic contributions can be generated by assigning constant, positive contributions to the atoms making up the investigated substructure and assigning a contribution of zero to all other atoms. To quantify how well the atomic contribution calculations can identify the ground truth contributions, we calculated the attribution ROC-AUC between the ground truth atomic labels and the atomic contribution calculated using our GPR-MGK models for each molecule. We then averaged these ROC-AUC values across all molecules in the test set. We observed strong performance of our GPR-MGK atomic attribution method, with attribution ROC-AUC exceeding 0.9 for three of the four investigated logic benchmark data sets. Only logic10 showed a lower performance with an attribution ROC-AUC never exceeding 0.72, indicating that it is more complex to correctly assign attribution scores to multiple functional groups.

To contextualize our attribution performance, we compared our achieved attribution ROC-AUC values against the attribution ROC-AUC of explainable GNNs. Sanchez-Lengeling et al. investigated the atomic attribution performance of four GNNs using six attribution methods, leading to a total of

Table 2. Attribution ROC-AUC and Relative Attribution ROC-AUC of GPR-MGK and Four Best GNN-Explanation Combinations^a

methods	logic7	logic8	logic10	benzene	liver	Ames	average
attribution ROC-AUC							
GPR-MGK	0.90 ± 0.03	0.96 ± 0.01	0.72 ± 0.04	0.99 ± 0.00	0.64 ± 0.04	0.81 ± 0.03	0.84
MPNN-CAM	0.80 ± 0.04	0.95 ± 0.02	0.72 ± 0.03	0.99 ± 0.01	0.64 ± 0.05	0.71 ± 0.03	0.80
GCN-CAM	0.81 ± 0.06	0.89 ± 0.01	0.62 ± 0.02	0.99 ± 0.00	0.58 ± 0.06	0.72 ± 0.03	0.77
GAT-CAM	0.66 ± 0.05	0.86 ± 0.02	0.69 ± 0.03	0.98 ± 0.01	0.55 ± 0.05	0.68 ± 0.04	0.74
GCN-IG	0.72 ± 0.07	0.68 ± 0.03	0.59 ± 0.07	0.94 ± 0.06	0.63 ± 0.08	0.56 ± 0.05	0.69
relative attribution ROC-AUC [%]							
GPR-MGK	100	100	100	100	99	100	100
MPNN-CAM	88	99	99	100	100	87	96
GCN-CAM	89	93	85	100	90	88	91
GAT-CAM	80	90	96	99	86	84	88
GCN-IG	74	71	81	94	97	68	82

^aBest performing model per data set (column) is shown in bold (p -value < 0.05, two-sample t -test).

24 combinations.²⁴ The attribution ROC-AUC values of our GPR-MGK and the top four GNN-explanation combinations from Sanchez-Lengeling et al. are listed in Table 2. For the logic7 and Ames mutagenicity data sets, GPR-MGK achieves the highest attribution ROC-AUC and significantly outperforms all four GNN models (p -value < 0.001 for logic7 data set, p -value < 10^{-7} for Ames mutagenicity data set, Welch's t -test). For the logic8, logic10, benzene, and liver toxicity data sets, GPR-MGK performs comparably to the best GNN-explanation combination MPNN-CAM (p -value > 0.05, Welch's t -test). It is important to note that the liver data set is considerably smaller than the other data sets, leading to worse predictive and attribution performance for both GPR-MGK and GNNs. On the other data sets, attribution performance for different architectures varies, most likely driven by data set characteristics. Nevertheless, on average, the attribution ROC-AUC of the GPR-MGK approach is higher than that of any of the four GNN-explanation combinations and never performs worse than 99% of the best performing method (Table 2), suggesting that GPR-MGP provides an overall robust approach for interpretable molecular machine learning.

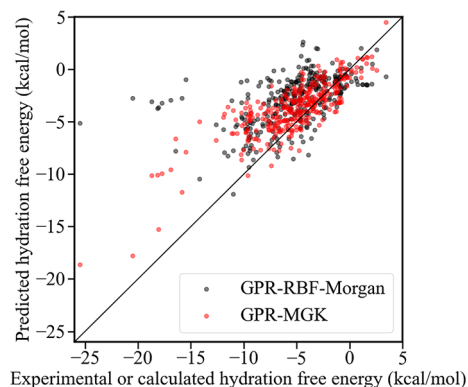


Figure 1. Correlation between the machine learning predictions and the ground truth of the FreeSolv data set.

Molecular Attribution: FreeSolv Data Set. We next investigated the performance of our molecular attribution algorithm that identifies the most important training data points to achieve a certain prediction. We chose the FreeSolv data set because it is an established benchmarking data set in molecular machine learning, and previous studies have shown that graph-based models significantly outperform classic models using Morgan fingerprints on this data.^{12,13,41} We again split the FreeSolv data into training and testing data sets

based on scaffold groups. Our GPR-MGK model achieved high predictive performance (RMSE = 2.75) on the FreeSolv data set. To confirm previous results, we also trained a GPR model using an RBF kernel on Morgan fingerprints (2048 bits, radius 2) and observed an RMSE of 4.07 for the GPR-RBF-Morgan model. GPR-MGK outperforms GPR-RBF-Morgan, especially for molecules with high negative solvation free energies (Figure 1), which is consistent with previous studies.^{12,13,41}

To understand why GPR-RBF-Morgan fails to accurately learn the FreeSolv data set, we investigated the molecular attribution for β -glucose (the leftmost point in Figure 1) because the prediction error by GPR-RBF-Morgan is largest for this molecule. The molecular structure of β -glucose and percentage contribution versus kernel value relationship of GPR-MGK and GPR-RBF-Morgan are shown in Figure 2. GPR-MGK considers sorbitol as the most similar molecule in the training set with a high percentage contribution of 0.72 (Z-score = 12.6, Figure 2B and Figure S3). This high similarity might be consistent with human chemical intuition because the two molecules share a similar atomic composition and differ mainly in terms of the ring opening of the cyclic β -glucose toward the open chain sorbitol. However, GPR-RBF-Morgan fails to capture the similarity between them with a percentage contribution by sorbitol of only 0.02 (Z-score = 0.55). Conversely, GPR-RBF-Morgan considers 1-bromo-2-chloroethane as the top contributing molecule with a percentage contribution of 0.34 (Z-score = 8.9, Figure 2C and Figure S4), which is surprising given the differences in chemical structure and atomic composition for these two molecules. To further investigate this effect, we compared the Morgan fingerprint bit vectors of β -glucose and 1-bromo-2-chloroethane and found they have a total of 23 nonzero bits and two shared nonzero bits (Tanimoto similarity of $2/23 \approx 0.09$): one for the secondary carbons and one based on a bit collision (defined as different substructures using the same bit position due to the hashing of molecular fingerprints to make them amenable to standard machine learning algorithms). Therefore, especially for such small structures, a basic similarity such as shared carbon atoms and problematic bit collisions in Morgan fingerprints may lead to an erroneous overestimation of the similarity of two molecules and adversely impact the molecular contribution and performance of machine learning models based on classic molecular fingerprints. Similar results for the other four molecules with the highest predictive error (the four leftmost points excluding β -glucose in Figure 1) are shown in Figures S5–S12.

When investigating the average percentage contribution spectrum of all 321 molecules in the test set (Figure 3), we

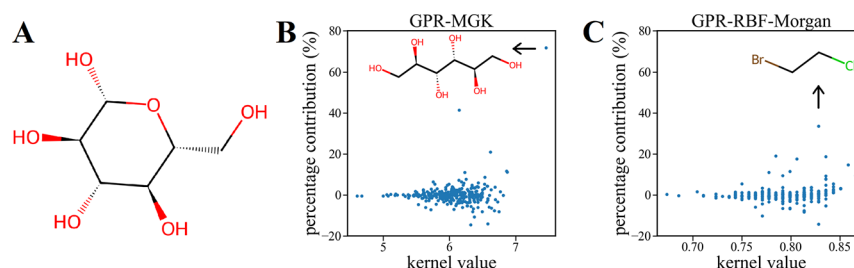


Figure 2. (A) Chemical structure of β -glucose, the molecule subject to prediction and interpretation. (B) Percentage contribution of the training data molecules set versus kernel value (similarity) for β -glucose using GPR-MGK. The chemical structure of the molecule with the highest contribution (sorbitol) is shown. (C) Percentage contribution of the training data molecules set versus kernel value (similarity) for β -glucose using GPR-RBF-Morgan. The chemical structure of the molecule with highest contribution (1-bromo-2-chloroethane) is shown.

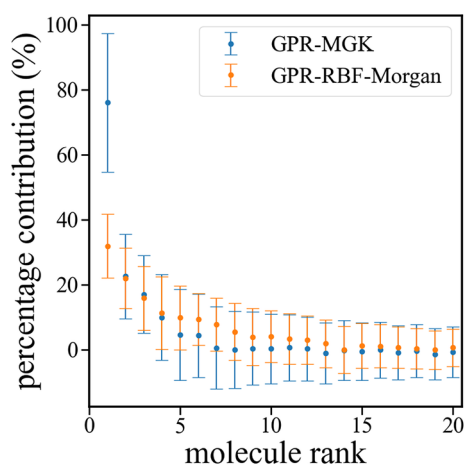


Figure 3. Mean and standard deviation of molecular contribution for the top 20 ranked molecules based on GPR-MGK (blue) and GPR-RBF-Morgan (orange) for all molecules in the FreeSolv test data set. GPR-MGK achieves significantly higher attribution of the nearest neighbor molecule compared to GPR-RBF-Morgan.

observed that for the top contributing molecule, the percentage contribution of GPR-MGK is 0.76 ± 0.21 , which is significantly greater than that of the GPR-RBF-Morgan 0.32 ± 0.10 (p -value = 10^{-116} , paired t -test). This indicates that GPR-MGK can identify statistically more significant relationships between training and testing data than GPR-RBF-Morgan.

IV. CONCLUSION

Taken together, we have reported here the mathematical derivation, implementation, and validation of two interpretability measures for molecular machine learning algorithms based on the newly developed GPR-MGK models. To the best of our knowledge, this study is the first work to introduce interpretability measures for GPR-MGK. Our results show that atomic contributions of GPR-MGK outperform state-of-the-art GNN models for established logic and real-world data sets to identify the most important chemical substructures and atoms to acquire a certain prediction in terms of both higher and more robust performance. Conversely, the derivation of molecular attribution enables identification of the most relevant molecules in the training data responsible for a specific prediction. Based on our analysis, GPR-MGK can identify statistically more significant relationships in the data than established GPR-RBF-Morgan algorithms, with a higher signal-to-noise ratio and improved weighting of molecular similarity to comply with human chemical intuition. Overall, our work further enriches powerful GPR-MGK models by providing additive molecular and atomic interpretability. We expect our work and extensions thereof to provide additional insights into molecular property prediction for chemical sciences and drug discovery communities.

■ ASSOCIATED CONTENT

Data Availability Statement

The marginalized graph kernel was computed using <https://gitlab.com/xiangyan93/graphdot/-/tree/feature/xy>. The interpretation of GPR-MGK was performed using <https://github.com/RekerLab/mgktools>. The data sets and code to reproduce this study are available at <https://github.com/RekerLab/graph-attribution>.

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.3c00396>.

Molecular attribution for trivial Gaussian process regression (Supplementary discussion 1); an example of a GPR-MGK model that achieves high predictive accuracy but low attribution accuracy (Supplementary discussion 2, Figure S2); the prediction performance of GPR-MGK and GNNs (Table S1); the computation time of GPR-MGK and GNNs (Table S2); the prediction and attribution performance for different hyperparameter optimization strategies (Figure S1); elaborated results of molecular attribution (Figures S3–S12) (PDF)

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<https://pubs.acs.org/10.1021/acs.jcim.3c00396>

Author Contributions

Y.X., Y.T., G.L., and D.R. conceived the study. Y.X. and Y.T. implemented code. Y.X. and D.R. designed experiments, analyzed data, and wrote the manuscript. All authors read and approved the final manuscript.

Notes

The authors declare the following competing financial interest(s): D.R. acts as a consultant to the pharmaceutical and biotechnology industry.

■ ACKNOWLEDGMENTS

We are grateful to the Duke Science & Technology initiative for funding. The computations in this paper were run on the Duke Compute Cluster.

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